Ronald Sladky ~ Active Inference Insights 012 ~ Amygdala, Priors, fMRI

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hello everyone um I just wanted to jump on before the podcast starts just to give you a prior warning that unfortunately for a reason that I haven't been able to discern Ronnie's video is a little jumpy it's not particularly smooth so you would notice that anyway but I wanted to yep give you a heads up obviously if you find it too much don't watch the video but it's a wonderful conversation so please don't miss out uh and yeah you'll enjoy whether you're watching it or listening to it but yeah just wanted to to let you know okay enjoy the episode hello everyone and welcome back to active inference insights I'm your host as always darus parvey Wayne and today I'm thrilled to be able to speak to Ronald sladkey Ronnie is a cognitive neuroscientist at the University of Vienna Austria where he teaches courses on cognitive science and predictive processing his research focuses on the amydala as well as emotion processing and social cognition in the human brain brain in addition to this he has been part of a team developing realtime fmri or functional magnetic resonance imaging where people can learn to regulate their own brain States while they are inside the MRI scanner Ronnie welcome to the show thank you so much for joining me it's it's uh really exciting to have a proper full fully fledged neuroscientist on the show thank you it's really a pleasure to be here so you you know part of the reason why I wanted to have you on is because you wrote this fascinating paper grounding what you called the amigdala complex in active inference uh and I have plenty of questions about that but before we actually go there I thought an interesting starting point is that you one very rarely meets people who teach active inference and predictive processing um at least in my experience within universities it's kind of like oh there's this little thing that started emerging and there are these kind of um esoteric people studying but let's not worry too much about that let's talk about reinforcement learning or or something a little bit more accepted what's the reception being like teaching predictive processing and active inference um within your courses okay that's that's a good question I mean I think uh it only works uh because of the master program that we have in Vienna so we have an inter disciplinary master program in cognitive science so I was one of the first students actually study this back in 2007 I think and so it's located at the philosophy faculty but it's interdisciplinary so we have some kind of core seminars and then are sent out to all the different institutes say Neuroscience or behavioral biology and we just take lectures there and do our exams there and then come back to the philosophy uh department and then try to integrate everything we've heard so there has always been this kind of curiosity to say okay um we take up students coming from different backgrounds with different Bachelor degrees so my bachelor is in was in computer science um um and then we basically want them to reflect on the paradigms that they are using so um back then uh the most recent Paradigm was actually for E cognition and and from there we kind of moved on and so they asked me to uh teach now the uh new trends in cognitive science uh um um seminar and this was then about predictive processing fantastic what um around what year did they ask you to do that so I mean I started with let's say predictive processing in about 2009 or something so actually was already inspired early on after reading uh on intelligence by Jeff Hawkins so I stumbled on this book it was yeah a little bit outside the typical Canon on incognitive science but I stumbled on it and I thought this idea very interesting this idea of predictive coding and um then yeah um the interest actually grew in 2009 2010 and uh I was kind of familiar with the early uh writings of Carl Christen and from there on I started to tell people hey guys you know this is I think the next big Paradigm and at some point I was able to convince people also in Vienna to give me a course on this topic and it grew from there and every year I've got two 25 new students and teach them processing that's fantastic uh that's super cool I mean I was at UCL and again I'm not going to bad mouth UCL they gave me a very good education this was for my masters uh as I said interestingly although Carl's there and Chris fiff and all of these people who are very seminal in the creation of active inference very little so I think that's really encouraging that you guys are are teaching it um that said obviously there are some fantastic resources out there online whether it's the textbook whether it's uh something like Ryan Smith's ch tutorial IE but that's how have you seen the I'm curious about how you've seen the kind of transformation in the acceptance of active inference as an idea because I presume back in sort of 2009 2010 when people were still talking about feed forward models um you know it was probably frowned upon that the brain was constructing its experience and there was a so-called controlled hallucination going on so how you know what was that process like of people coming on coming on side oh absolutely I mean I was somehow caught between the lines so I mean I was coming from my cognitive Science Background to say okay let's um you know uh get over with behaviorism and cognitivism even and and you know uh move towards 4 e cognition but um at the same times I I had a very hard time to actually apply this to my work done in cognitive Neuroscience because after my master uh program I did a PhD in medical physics at the Medical University I was working together with physicist we uh mostly focused in uh um neuroimaging but also uh Dynamic caal modeling then and uh but they didn't really care for these ideas from cognitive science I mean those 4E models that we tried to come up with back then quite honestly they were very uh uh useless for cogn neurs and um so they always knew okay I'm the guy who knows how to code and because of my computer science background I also worked in the industry they knew that they could give me some data and then I will make sense of it and this is how how how they basically um yeah um handled me uh but they knew that I had this kind of esoteric side coming from cognitive science you know always questioning the paradigms and and and um not thinking that the brain is just reactive right then at one point I came up with um with uh my interest in in predictive processing and finally I mean the big confirmation was when when people uh like Carl and Andy Clark also started talking about it then people uh realized oh okay it's not only Ronnie saying something like this about productive processing it's also other people and this is how it got a little bit more accepted but honestly there were this basically two camps in in Vienna one very much into this kind of neobehaviorism and the other one more for E inspired in cognitive science and I was caught in the middle and now I I try to you know bring those camps together interesting okay that's cool yeah so that must have been what sort of 2013 2014 when the you know Andy clarks's book came out and whatever next was published and I guess that's when all the buzz around predictive processing emerged that's that's interesting you know it's I love talking about the actual ideas but I'm also fascinated in the kind of meta aspect of people fighting over their ideas um so so yeah so we had sort of fory and then then very strict computational accounts and then act of inference just sticking with for and act of inference because I feel like those are the two that are kind of spearheading cognitive science at the moment do you sense that they're compatible um there's been a there's been a real tension I would say between sort of the traditional an activist cognition side and autopoetic side and more let's say representation forms of active inference where do you kind of see the middle ground there and and their Confluence yeah I think they're totally compatible I because I mean like I said I'm I'm coming from uh autois and and this was was was the big the big motivation actually to seek out a kind of model that somehow um yeah accommodates uh the the ideas of autop prasis and so it was only natural for me um to stumble on the free energy principle at some point and say yeah this is this is how I think we can interface those those ideas to basically have something tangible to have something that can make experimental predictions um in terms of you know how you interpret behavioral data but also neuroimaging data and so on but the same time in its foundation is already also integrated in the ideas of uh autop poesis and um you self-regulation so this was already f for me from the start and now what I learn is that there are some semantic differences um I would say where people say no no no this is not the kind of an activism and body cognition that we uh want to have and um yeah I mean for me those differences are not so severe I would say I I would like to leave this uh debate to philosophers where you say okay this is maybe it's just my too sloppy reading of of the material but um in his foundation I think it's totally compatible and I think if we work together I and get the semantics right and and what we actually want to achieve I think we can come up with a very nice unified model to uh bring the whole cognitive science together fantastic I love the optimism I share it uh because I I also love the earlier work of verella and rash and Thompson and then you know vel's later work in the 90s actually trying to bridge the phenomenology with the neurology and the neuroscience and I think all of that stuff is fascinating uh orbe it incomplete that that as you say like we're still working towards those generative passages but um excellent and and it's funny that you did physics as well right because then you fold in the kind of so-called high road to active inference and the basian mechanics and the [**] plank equation and the longans was it cool to see all of that come into the picture given your mathematical background so I have to say I'm not a real physicist I worked with physicists so which means that all the ideas that I had I always discussed in the in our coffee room and they always feedback on it so it had to make sense in the physical world but this doesn't necessarily mean that I understand what the [**] plank equation is really about right so um yeah I have to admit um this is not my my uh my biggest uh expertise in the the field fair enough fair enough me neither so let's not even go there um I wanted to ask again before jumping onto the amigdala um I thought this realtime fmri idea is fascinating um and I was really I guess it's not it doesn't need to be too related to active inference I guess the nice thing about this podcast is we can just as I said before we went on go wherever um how does that work and why was real time fmri a historical challenge was it just people just didn't think about it or did it prove uh challenging to um predecessors so I mean first of all it takes quite some uh some time to to process the FM data so um uh I I think it goes back to like 2006 where where the first people started to do real time fmri um and back then there were many pragmatic challenges back then so how to get the images quickly off the scanner onto some computer to do the analysis and um these things have um you have been res solved over the next couple of years so the scanner manufacturers were more accommodating to the researchers needs and this was this really improved the field and then also uh data analysis and pre-processing um was was was very much improved by new methods to do that really in real time I mean one fundamental problem that still exists is that there is this hemodynamic DeLay So you always when you do functional MRI um you don't measure neural activity directly you measure the blood flow in the brain which means there's always a delay by um about six seconds uh when you have the peak uh of the hemic response and some people said for philosophical reasons that this is much too late actually for somebody to interfere so if you want to you know regulate your own brain then you will always see the uh the um the e um the outcome of this regulation effort about 6 seconds uh later and so some people say okay I mean this means by principle real time FM is impossible for for neuro feedback but um I disagree I think the um humans are very good in association learning I mean they can uh um they can steer huge trucks you know trucks much larger than they are they can huge carrier ships whatsoever so um this is what you can basically learn by getting used uh to the uh to the task and um and and yeah getting getting a little bit of training and when I wanted to do when I was when I was studing my PhD uh uh program actually so I had a job interview by my supervisor Kristen wesburger real hardcore fmri physicist uh from from the early days of FM actually and he asked me okay so Ronnie what would you like to do after all so what what is your career goal and said oh I really would like to learn this real time from Ry and back then he said no way this is not going to work no there's so many technical challenges and um first you should really learn your proper FM and also a little bit Dynamic Cal modeling uh which was quite on walk back uh back then it came up as a new uh method and this was absolutely the right choice so I didn't waste any time uh during my PhD to you know uh you know spend endless uh uh time in the lab to really set up the those those pipelines so I waited a couple of years for my first uh uh post abroad um in in surich at the bule Clinic um where I had the opportunity to work with Fran chowski was also at UCL um at the welcome trust Center where he also was one of the pioneers of of real time fmr and this is where we started and this is how we um got it going and we set up um a small uh pilot uh experiment I would call it so with different interventions people had to regulate their own Amala so I'm I'm always a little bit Centric uh we instruct the people inside the scanner to either up or down regulate Thea and we used uh emotional faces as feedback so one instruction was for example to make um neutral faces happy so we used morph faces and then they looked more and more happy and uh or to downregulate the amula by making uh fearful faces less fearful m cool and so so it's you know people it's difficult you know as you know as an experimentalist it's hard sometimes to understand exactly what your participants are doing in a brain scan or when they're doing an experiment I presume you was asking them question as as quality to like what tactic are you using because you've got this sixc delay but then you've also got this functional like oh you've got like let's see how you're actually actively um modulating the response of your amigdala what kind of reasons did people come up with even if they're kind of spurious and and post hog yeah uh I mean we uh we didn't have a systematic way of asking those questions but I wanted to um do some qu qualitative analysis of of the data and you know just to collect some samples of what people are reporting what kind of strategies they might be using or not and so I always asked them how certain they were that they were performing well and um so this was yeah not not really indicative of their actual performance so I I remember this one particular participant she was absolutely stunning she was really able to up or down regulate Thea at will you could really see how the signals would go up would go down again so it really like this textbook example what you show in a post is a typical subject and then we asked her so how satisfied were you with your regulation and she said oh no it could have been better and I'm not sure and um when you also ask him for strategies what kind of um strategies they used it was also very hard to pinpoint because some people had really uh good ideas so some people had experience with meditation some um some were maybe a little bit from a psychology background and had some training uh in in Psychotherapy and they did let's say um um different cognitive regulation mechanisms that a priori would make lots of sense but they failed on the other hand we had some some guys from um let's say from the eth so from Technical University there and uh they said yeah without knowing anything about the brain actually said yeah I was just lying there and thinking about playing basketball with this person I'm seeing and reason so so the only the only thing that um that seemed to work was to consistently pay attention to uh what you're doing so if you're trying too hard if you you know try to adapt a very complex regulation strategy or something um and you don't manage to focus your attention for the whole time then it will not work but if you're very consistent with your strategy I think that brain can also take care of the rest maybe oh this is okay this is really cool this is really interesting I'd love to go a little bit further into this um so just so I can get some sort of clarity on the Paradigm um you're showing people these faces some of which are you're expecting to raise igdal activity and others you're expecting to sort of downregulate igd activity and so where does the self-regulation aspect come in and um they're not party to the actual the fmr information of their amigdala is that right um it is so is basically so they can see what's they can kind of see what going on and so why would so my question there is why would that person not think they're doing very well if they can see that they sort of conscious well again philosophically Sticky words here but like their will their activity their attention is actually having this modulatory effect so why would they have this like uh this kind of self-deprecating self reflexivity yeah it it might be hard to really quantify how uh and um to really know how good good you are in in relationship to to um um to other people baby maybe because um uh yeah they don't they don't have much experience with that and they might be a little dissatisfied if it takes longer than expected if you know they try to focus but they also see the signal going in the other direction for a brief moment this so it's very hard to tell actually okay so people just have very high standards um so so okay so let's let's just get some clarity on the Paradigm so we've got these faces different faces the so that's going to kind of um exogenously affect the amigdala but then you've got you've been telling these participants okay we want to see if you can contribute to that process so let's say I'm looking at a happy face um what would you be saying to the participant in that sort of context and what effects would you be hypothesizing that you might see so what we know from the literature is that mucular activity scales with the intensity of the emotions that someone is is perceiving and um so what you see is there is already some kind of uh um uh dependency of the stimulus that is shown and what how the aric should response so in a way what we talk about here is a closed loop between um between the stimulus uh that is also uh um uh depending on the activity and uh you as the agent inside the MRI scanner are somehow in the position to manipulate in that so find a way to to influence this closed loop between what the amula is doing in processing the faces and what is actually presented and uh I mean this was without much Theory actually so there are are control theoretical assumptions on that that you're somehow regulating uh this uh disclosed Loop but um at some at some point we could also come up with an active inference mod lecture what might be going on there right yeah yeah it definitely lends itself to a kind of hierarchical modeling perspective where you have kind of higher order constraints I guess it was the uh was the role of the kind of endogenous attention or endogenous will to always like oppose the closed loop so if for example I'm seeing a very fearful face so we're expecting this to correlate with increased the migler activity would it always would the task always be well try and deregulate or or down regulate the activity of and maybe if it was again I'm just asking I wasn't there uh but if uh if there sort of minimal activity because you're just see a very neutral face you might say well try and try and Spike it was it always kind of doing the inverse of what might be happening endogenously no we actually did several interventions with that what we found what was working best was actually to either downregulate the to make fearful faces less fearful and um neutral faces more happy by upregulating DEA so if it was um somehow consistent with uh with this natural tendencies um then uh then it it work best I would say wow this is I really like this this is cool so um as you say it kind of lends itself to this active inference hierarchical perspective I believe you did I know this is a weird thing but obviously I search up everyone I'm speaking to just so I can do my intro and stuff and I think I saw that you did some work on shophow um I it was a long time ago it was your Master's work right that's right yeah shop shophow does you know shophow focuses very much on the will what I mean this question just came to me so it sounds very contrived and but it really isn't um what role would you say this points you know for a kind of naive audience or folk psychology audience it might say well of course this is just the will this is me doing it this is like purposeful activity and I can control my thoughts and I can control my actions and so on what what do you see being the kind of uh operating um function or operating mechanism in play here in terms of top down regulation yeah that's interesting I mean um so when it comes to the brain now um I would say it has something to do with prefrontal control of thexa uh if we talk about yeah more um metaphysical accounts What Chow could give with with metaphysical will I mean a short explanation first because this is yeah as you said this was my master thesis and I stumbled in schopenhauer while um reading a little bit of uh philosophical literature in my third and fourth semester in my Congress science studies so wienstein but also schopenhauer then I was really fascinated by how many interesting ideas he already had in the um 19th century and I thought that many of the things that he was talking about are still state-of-the-art discussions in cognitive science and that's why I I tried to you know apply some of his ideas to um to current problems that we have that's why I called my master thesis schopenhauer's last will and um it's not very good philosophically speaking because I'm not a trained philosopher but it was um it was very interesting actually to uh to step into this debate and discuss with philosophers about it and this was was really informative but this was kind of a a singular point in my uh history before I moved on again to the more empirical more uh uh cognitive Neuroscience side of cognitive science but let's talk about the will now and the way I interpret shophow so I mean at one point I really want to take a philosopher and write a proper paper about it but I think chenhow had lots of interesting things to say um that are still relevant for active infrance as well because he talked about two things about representations and will the world is Will and representation and representations um have a lot to do with the models that we are using uh in in cognitive science so he didn't use of course representation this loaded fashion that um that we now have in cogntive science where the representation Wars um but uh it was the the the the the German word of representation would be forell which is like an imagination so there is some kind of um uh imagination going on that I used to actually uh model the world so we cannot access uh access the world directly so he was very much influenced by c um there's only the only the consequences the phenomena are somehow uh reachable to us and based on that we form imaginations representations of the world and for him of course this was not enough and I mean when we now translate it also to predictive process and we see okay just you know s yeah collecting samples of the world and compiling it somehow this would be very passive uh predictive coding without any intention without life it is just a way of of storing data and so so shopen how already needed to have this kind of second uh uh Power at work and this would be the will and the will was nothing um meta um metaphysically rich in the sense that there was some uh God and its will and uh it has some some yeah it was more a force of nature very much undirected so he would say on the most primitive level um it's it's the laws of nature uh that are a manifestation of this will and then when you go up uh on uh on living in in the Hur of living systems he had this very strong hierarchical view you see that plants somehow manage to fight against gravity because they're growing upwards and and you you get from there to animals and then uh human beings but in essence they are a manifestation of the same will of the same um force of life or maybe of the same principles that we have in nature when we think of thermodynamics for example we need to find a way to fight the second law of Thermodynamics right and these mechanisms um could uh could be yeah could link to this merry metaphorical idea of will interesting cool yeah yeah yeah I mean obviously yeah the notion of a will especially of a sixc delay is uh is tricky right um philosophically but I think it's it's not a it's not a it's not an it's not a redundant question some philos you know the really strict materialists might just say well it is just the prefrontal cortex uh but at least on this podcast I don't hold any ontological requirements so if you wanted to say that it's a pan psychist Proto agent or if you wanted to say it's God I'm I'm I'm I'm gonna entertain it but at least in terms of what I would like to say to be a little bit more precise about my idea of of free will is not necessarily you know philosophers somehow like to uh detach the actual problem of living systems to very abstract spaces where time doesn't play a role and for me six seconds delay are um is not really a problem because I think uh that uh agency has no nothing to do with uh you know moment to on the level of moment to moment but also in the level of process that you are causing something you you see the outcome of your actions you might not be even aware of uh um of of the behavior that uh gave rise to these outcomes but then you evaluate the outcomes and then you learn from it and then you adjust your behavior I think uh in in this kind of ballp part this is where most of free will actually happens because most of the behavior that we do is automatic and and for good reasons because it's efficient and fast but over your continuous uh effort you're able to uh steer um yourself also when you have very delay feedback like in Neo feedback yes yes I think I think a lot of these sort of broader philosophical questions are underpinned by your like resting Axiom if you're taking the notion of a substantial self a cartisian dualistic self then sure like it it the time probably doesn't matter right there might just be a delay but if you're just taking this neurophys phological mechanisms well then we're not going to get off the ground by even talking about you recognizing the consequences of your own action I think it's funny because shophow also said I believe uh a man can do what he Wills but he can't will what he Wills which I just a beautiful reputation to kind of soft determinism I always thought was the best reputation of soft determinism which is like yeah yeah yeah like I'm doing what I want but I'm like well why did you choose this that you know why did you choose what you wanted and you run that infinite regret he he was a brilliant writer he was really really great in um you know making really good points really uh um uh really uh high in rhetorics and everything uh but the point was I I think that he was misguided and in a couple of ways and I I think it has something to do with his lifestyle he was he was a very odd person he was living alone he didn't have um much social interactions and uh also he was very mean to friends and people interested in him so um yeah never a family never a relationship whatsoever so I think he has had a um very depressive uh deposition and um and I think so I mean this is what now now we move back into into into the clinical Neuroscience domain maybe when we look at people uh with uh depressive symptoms they also tend uh uh to be more um uh incompatibilism uh basically driven by external factors right so there could be this interesting relationship there as well oh yeah yeah go ahead go ahead please and so because I mean it was kind of a dead end for him life it was always uh was it was always governed by external factors and there was no way out and um this it just drove him to this extremely strong nihilism and also his uh idea of anti-natalism actually it's better to um not have any you know offsprings at all and to die out at one point so it's very dark very um uh um nihilistic and this is definitely the side that I don't like about it yeah also here disregard for women yeah well they were all pretty rubbish in that domain I think you know um yeah I think we can kind of we can in some sense those elements of those philosophers um the disparaging attitudes towards women just kind of that element Dooms them all uh I'm not sure shophow is an exemption that I mean of course there are some who were a bit nicer but I think they were all pretty twatty in that domain so that's not great um that's really interesting yeah you you hear this stuff about sort of locust of control internal versus external locus of control and it correlates with um meaning and lies and and depression and stuff like that I think what's interesting there is even to make that point just on a philosophical basis you're you you're still you're still investing in this kind of physicalist Axiom which I just don't think is particularly well substantiated by science or by philosophy or by whatever it you know this notion that well we we just don't know um because we have this remarkable ability to be subjectively aware of things and have intentional stuff answers on things and well Ronnie if you can find a person who can explain that to me then you win right and I'll become shophow and nihilistic but until then I'm quite happy to sort of bathe in the the loveliness of that mystery and I wonder whether if shophow had read Thomas Nagel or David Charmers whether he would have been a bit more optimistic but you know I think you sense a lot of this in this the scientific turn in the 19th century uh yeah and the 20th century this this gross nihilism which obviously n basically foretold more than anyone so this is I mean so I think this your your integration here of the therapeutic landscape is interesting as well because you know this is very broad brush and rough but a lot of therapy especially for something like an anxiety disorder will be will look something like reestablishing the prefrontal cortex as the master over the amigdala so I'm worried that the world is going to fall down or plane is going to crash or well like you know this why people like Vagas nerve breathing but let's reestablish the role of the prefrontal cortex to moderate and regulate uh that overactivity but it almost sounds like okay if we focus so much on that neuroanatomy we almost lead into an nihilism because well the prefrontal cortex isn't me right if you showed me my little flesh of prefrontal Cortex I wouldn't say oh well that's my will that's my desire that's my capacity for regulation so what was your experience when you saw that people were able to regulate their amydala regulate their emotions was it liberating for them or were they kind of spooked out so yeah that's that's an interesting point um so yeah because this is actually where where I'm also coming from I always thought of the being regulated by the prefrontal cortex so uh uh this was also my main focus in my PhD thesis worked with people with social anxiety disorders and um you could see in neurotypical people that uh when they were watching emotional faces they were able to down regulate the amula um by uh by by a small region called the medial orbital frontal cortex and precisely this mechanism was not functioning so well in people with social anxieties order and in another study where people were uh given uh ssris so anti % uh medication uh they were able to have stronger down regulation of the amular or with the frontal cortex so this very much played into uh uh into this kind of narrative and this is also what we find in the neuro feedback study that uh people who regulate the Amala also have stronger influence of the am of the prefrontal cortex on the Amala so this kind of of works it it it um it has something uh also to do with with the uh studies on um uh cognitive uh effort and how people are doing cognitive reappraisal and then are able to uh regulate subcortical areas by using prefrontal areas so this kind of network but you're right this is not the full story yes well we can't we yeah we'll move on but I guess I'm I'm really just trying to Ram home we can't discard the phenomenology uh and as a fore man I'm sure you're you're you agree yeah but you see where my problem actually is so uh I I I kind of accepted this kind of narrative that we use in in clinical neuroscience and cognitive Neuroscience to uh Model A mucular connectivity right and um what I've been actually struggling with the um uh with the topic of amuka and active inance was for many years so like more than 10 years I was not able to B basically bring those two together I I that I was a mucular research uh Searcher during the week and on the weekend I could think about active inance but those were things that were highly compart comp compartmentalized I couldn't bring them together uh why because I mean it was you you could say okay I have to think about top down control of T MCA but like you said I was not happy just to say okay it should have something to do with the um with the prefrontal cortex and the prefrontal cortex regulating but yeah there's more to that you mentioned um breathing you basically a link also to uh inceptive processes will regulate also deala so if you are uh if you have a panic attack it has nothing to do with what you're thinking you're experiencing that you're losing control of your body and this alone is enough it has nothing to do with some kind of abstract reasoning that you're doing it is an embodied process that should play a role so um then the other thing is uh so it has something to do with the insular cortex then maybe or it has something to do also with uh brain stem inut into the am um the other thing is then also reward processing because um much of the debate is is focused on on uh fear conditioning experiments in animals because it works so nicely uh and this is where you say okay this is um this is the function of the amula it's about fear and threat processing or something and even in my um in my PhD thesis I wrote about also the U you know that nuclear all responds to positive emotions and I had a big fight with one of the uh reviews of my PhD thesis because I said no the is just for fear even though it already outdated this kind of few back then but yeah nowadays we know so much more from animal experiments that um the amula is is heavily connected with the vent stum so it has a lot to do with you know uh making the decisions about risk and rewards and balancing those and um so we now know from the animal models very precise mapping so those those circuits definitely exist somehow and so this is another eonr then so where should I start to actually uh draw the top level of the hierarchy you say okay this okay the amler is driven by reward no we could also say that no the am is driven by interception and uh therefore I kind of failed to come up with a way to start my active inurance model and this shifted then a little bit later when I had this inside for for the transy cognitive science paper yeah yeah we we'll certainly come to the text paper I guess one thing yeah one interesting addition to this uh conversation that the pre the the loop between the pref CeX and the amigdala is not enough is OCD and OCD is interesting because it was cast as an anxiety disorder and this you know and so you would say okay well cognitive behavioral therapy is about reinstating the role of the prefrontal cortex but OCD actually involves hyperactivity of the orbital frontal cortex and so and how that maps on to therape you know therapeutically is they'll say don't do loging don't do rumination don't do don't logic your way out of um a problem right that's actually worsening the problem so and and what's odd is that in many ways the symptoms of OCD are very similar to the sympt Syms of something like Gad or generalized anxiety disorder and yet it seems like there's the inverse relationship or the inverse problem where you have hyperactivity over the frontal cortex and so you have stuff like TMS or you know intrusive Lees going in there and actually trying to deregulate that process so it's um it's a mystery but as you say it's it of course it's a mystery because it's you know hundreds of millions of years old it would be lovely if it was just one Loop uh but alas okay so let's um let's jump to this tix paper um your Trends in cognitive science paper on the amigdala so you called it the amigdala complex which I really liked uh I thought it was kind of it sounded kind of Victorian um before you fold an act of inference into the amydala you mentioned that you you know one of your reviewers of your PhD went well amigdalas are only for fear what are the other kind of preconceptions that historically have held about the amydala I would say that there is only one amigdala but essentially what's what's very important is I mean of course there are two they are bilateral but um even more important the lateralization is that the amular consists of at least two completely different nuclei and the one would be the Central amca and the other one um the basolateral complex so these are actually two different um nuclei that somehow are fused together or um over time and the origin of the centrala is actually the the vent strum and while the uh basal and lateral part of the amula actually more cortical Lian structure so this was this this is an important game Cher already I mean many people in the um in the rodent literature know about this I mean they M the differences between central lular medial lulara all the different parts of the basolateral uh complex and they're totally familiar with that but in human Neuro Imaging we mostly focus on the whole amuka or just some active voxels within the gross neighborhood of where Thea might be or something so it's it's um it's very unclear when we look at the neuroimaging uh data what we actually mean when we talk about Thea because uh many groups uh talking about different things and it may be worth just unpicking for the non TR audience when you say so the central mdala um well actually I guess people are coming from an active inference hierarchy so when I was reading I was thinking okay the central migler is higher up the hierarchy uh and provides predictions to the basil lateral area but when you say okay emerges from the vental striatum so it has something to do with motivation and reward and the basolateral feeds back into the sensory cortex if we now feed in these classic terms of perception and action which are kind of the currency of active inference what does that say more broadly about these different bits if that's a feasible distinction okay let's let's start maybe with um with the different historical views so let's say um the simple human neuroimaging view might be the Amala is something like a fear detector or Emotion Detector or relevance detector something something like this and if uh let's say a a new emotional phis pops up then the would increase its activity this is the a traditional few in neurom imaging and many experiments are done this way and there yeah some really good experiments and you can learn a lot um from this kind of mular re activity the more nuanced fuse nowadays in Neuroscience would be to say okay let's look at the different sub nucle um let's say in a mouse model for example to say okay we have the Bas latal Mula at one end at the lateral end of the Bas lateral Mula you have unimodal inputs and those are integrated towards um the more basal part and become a multimodal uh um uh in integrated uh stimuli and uh from there there's an um there's a connection to the central amula and the central am would then be the output region so there's some kind of gating that is modulated by let's say ventrum or prefrontal cortex or whatever and um there basically decisions are made if uh you should show some kind of stereotypical uh threat response like fining Behavior or fighting or running away so it um very much um depends on the context what kind of behavior is actually uh used to get away from this kind of situation and this is the mainstream view yeah what I tried to do is then to to basically flip it around to say no Central amula is is not uh um just receiving the uh the signal but it is actually the one that is on top of the hierarchy and sending down predictions to the Bas Metro to that and do you so yeah that's great so do you then uh reconfigure or recast this gating mechanism as a contextualization mechanism would that be fair would that be fair to say I mean so many of the things that we are seeing in this paper are very compatible with what people already know so many people who follow the classical feed forward perspective would say yeah of course what uh uh the output of the central amula is additionally gated by input from insula PFC ventum and potentially also other areas and um so this already um already is there but for me the big question was actually but how does the basal amular know what kind of data to forward to the central amular so what is Rel for the for the centrala and this big question of meaning and relevance this is something that is not answered in the classical feed forward uh Paradigm this is what we had to change so it's worth pointing out there that the transmission of information from the unimodal to the multimodal to something like the abstract level at the central migdala is not necessarily it doesn't it doesn't it's not like a replication process there s like stripping out of information as it gets p up the hierarchy is that is that is that fair in terms of relevance meaning and yeah I mean personal relevance right it's it's going to be individual specific what do you therefore suggest that those nonrelevant stul like just get as we say explained away yeah exactly so it's it's a kind of abstraction gradiant as I like to call it so you you get away from very concrete uh sensory features more to something uh uh generalized uh that is maybe even detch charged from the context is uh uh longer also in time and and um so in the temporal Dynamics actually so you uh um so you explain smaller stimulus features to fluctuate more uh uh more quickly than um you know the object identity for example or the the effective state that you are in where you think that you are currently under threat or uh even that you're in a a constant state of anxiety for example yeah so so we can so just for like yeah just for myself and the audience it's kind of this idea that we have these slower contextual fluctuations which can regulate the faster fluctuations so maybe maybe an example would be worth using here so I I like I really like your example I actually used it as well and I genuinely did not take it I just uh it may have been something that Carl mentioned but it's the kind of snake on the lawn rope on the lawn dichotomy so you have a rope in your lawn and it's kind of curled up in the grass so you don't know whether it's a snake or not now your lower level parts of the amydala are just responding to the sensory input which is snake like right the curled the texture but then you have this higher order contextual prior that no you're in your garden right so you're in Australia I'm in London like we we don't have snakes what where do you find that um to meet and therefore the prediction error uh gets suppressed is that in the central mdore or is that higher up so I would say that um the central amula has more uh something to do with uh having a general feeling of how safe you are at the moment safe also in terms of how well you are regulated and how well you expect to be regulated in the future so if you experience experiencing pain or death in the near future then you're not saved and this is something you should be afraid of and this is where I think the centrala plays a role so if there is sufficient contextual evidence that uh okay you are in a tropical jungle and there might be snakes out there um then you're of course more on the edge so and this would mean that you're upregulating uh uh those parts of the amula the basal Milla that is very good in detecting those uh those snakes while when you are in a safe context when you are sailing this is the ex example that I'm I'm I'm using in the paper not you know that there many ropes lying around but there usually no snakes um then uh you should be actually quite at EAS and not misinterpret all coiled uh snake like objects as snakes yeah I think I think this is really I think this is really interesting I think it's an area that actually I've been talking to some people about this the fact that I don't think these attentional mechanisms are actually that well articulated in Act of inference literature there are some wonderful papers I keep pointing people in the p uh direction of miter and his papers on attention with Carl and others but I think this is a really interesting point because I view attention in some way as a uh as a mental action which involves priors and so for me I when I talk about the prior in terms of attention for me it's like okay we we recognize through these uh slower contextual cues where you are and that constitutes um sort of as you say there prior involved here in the down regulation of gain control or the down weighting and precision waiting over anything which might fit the template of a snake but then people I guess generally think of that in terms of the likelihood distribution right so what are the you know given my state what's the the likelihood of having this observation of a snake but for us we're kind of saying well actually this might be constrained by higher order uh action policy maybe it's worth unpicking because I think one reason I really really liked your paper and I think it's a really fascinating paper is because I think more than anyone I've actually ever read you unpicked the problem with the ambiguity of the term prior so you say there are two types of priors there's kind of the phenotypic prior or what you call the preferred prior I believe preference prior preference prior very good and then you talk about the prior's aiding perception so kind of the the priors I've been talking about here in terms of um suppressing prediction error um and then also belief updating so maybe it's worth putting on your teaching hat and just saying okay we can keep using the snake example for people who are very confused at this point because they're like why are we using the word prior for these things maybe it's worth you on helping us sort of unpick those two senses of the word prior MH okay so yeah I'm I'm sometimes very liberal with using the word prian you know this comes actually from this cognitive science debate of whether you have a model or you are a model and I think it's it's usually both so metaphorically we can say that our body and yeah the fact that our Body Works in this kind of environment that it is a very good model on on its environment and I think the same also applies for PRI I think you are able to um construct uh prior that you're able to um to uh introspectively assess and also report and do reasoning about it but I think that some priers are just let's say hardwired in a way I sometimes do U use this as an example uh for my students I would say that even the the bone structure the skull these are also prior in a way they're very inflexible highly precise prior but in a way they allow us the whole organism to do inference about the world right and then maybe C differentiating that from uh the second type where you say that these priors and here I can quote Aid perception by guiding attention enhancing the relevant signal features and suppressing noise in Sensations how so that's one side the homeostatic prize the preference prize these are somewhat fixed unless you're sleeping and you can let your body temperature go down but that as side they're somewhat fixed how what's the difference between those and these priors involved in perceptual belief updating right I mean I I always like to make this distinction um between uh predictive coding and um predictive processing or active inference so maybe some of my students are watching now and this is a typical example question I like to give for the mass defend you I'm so jealous of these these kids kids as well am I talking about but I'm so jealous great exam question so I mean difference is now perceptual prior can be seen in this uh very stupid uh operative coding way so you have just a system that just tries to absorb uh the environment as it is to minimize uh the the the difference between what is actually sensed and what it already knows so this is coding nothing really interesting so to speak from from the uh from from from from cognitive science or from this idea of organizing a living system I think the interesting point comes if you say no it's not only about updating uh uh your ban beliefs it's also about uh ban regulation and this is another formalism basically that you can easily introduce uh into the whole idea of uh of Bion processing by saying no some of the priers they don't update they are fixed so in terms of perception this is to give you a very simple example you might have the idea that there could be a snake out somewhere you take a look around and you don't see any snake so you update your prior belief okay well I heard something but there is no snake so this is okay so it's easy to update those very flexible priors um but there are some priors that have to be are more or less fixed and that's for example ideal uh body temperature or blood oxygenation for example right sometimes I hear as a criticism of people who are not familiar with predictive processing to say oh wait why don't you just update your prediction that you you'll die of course you die and this has something to do with uh um fixed preferences that you have and so there is no way of actually reducing prediction error and I think having prediction error on on on on some levels of of your uh of your hierarchy and this is something that we have to learn uh and figure out experimentally in the future where it actually where it actually plays the biggest role uh leads to the feeling of uh negative veilance so if there is prediction error that you cannot minimize it doesn't feel right there is something something very wrong about it so the only thing and this now closes the loop how you can actually uh minimize the prediction error between your Sensations that you're having for example to little oxygen and your homeostatic set points your preference p is by doing actions you can you know uh breathe again you can if you're too cold you can put on a jacket so um this is how action is the only way of actually making this right again yeah yeah and as I've said to my audience or many times this is perfectly cast within the variation of free energy um equation itself right reducing K Divergence recognition density in the true posterior then also maximizing model evidence I guess so two things come to my mind here the first is that we also talk about now we're using these pomdp schemes and Baye graphs we also talk about priors over um States or or or and priors over paths and prior over action so we're talking about you know preferences sensory prefer sensory observations of my action or we're talking about State Transitions and we're talking about e you know the eent so we're talking about habits so I guess the reason why I bring that up is because for me it's an incomplete picture without mentioning that because in many ways right I might for me I actually think it's important to divorce these different types of priors because without that I might say well I have an inflexible if I if I say that they're all one type I have this inflexible prior that I don't want to be around snakes but that means I get this perceptual information right with precise uh likelihood distribution that I'm probably you know this observation is quite likely given that there's a snake but I don't uh update that prior so I'm in real trouble right because there's a snake but I'm not updating it but I guess the point here is well if I recognize at the higher level of the hierarchy that I am in a jungle for example I can now start doing this mental action where I through my mental action increase gain control over the likelihood distribution the a matrix and downregulate my uh prior right that doesn't mean that I discard my homeostatic requirements it just means it's more likely that there's going to be a snake and I'm going to respond accordingly um I think a really useful piece of terminology here actually comes from Sam J nagoshi who who works with verses and is writing this textbook on basium mechanics because he talks about uh perceptual priors and what he calls autonomous priors and these autonom priors are these like genetically determined or or or very you know deep rooted learn cultural priors which as you say are super high precision and um not affected by the idiosyncratic context in which we find ourselves so maybe it's it's it's worth picking up here the role of attention uh I'm talking about the sort of hierarchy here the picture that I've laid out does that does that sound okay and does it kind of conflate issues by adding in like a third type of Prior which are the prior over action no I don't think so I I think it totally makes sense um so I mean the am uh cannot solve all the problems on its own so it has to be somehow integrated in that I mean this is this is what we uh how we also tried to reframe this whole idea of um uh of seeing the central amular is the only output region I think that every part of the amula has its output um that player relevance here in this kind of uh um uh uh perceptual and action hierarchy so the thing is when you um when you when you talk about attention I think that uh predictions from uh the uh lateral am to the unimodal cortices for example um they upregulate uh uh areas there to be more sensitive towards incoming stimuli so this is also a kind of um neural gain control has something to do with uh attentional processes uh this uh the same also for the central Amal it's not only for for actions but also uh regulating uh arousal through uh through the whole body so if it's uppr regulating arousal um then uh your whole uh physiology changes your vision the way you sample the world totally changes is a top down phenomenon your um um your way of how you um yeah integrate interceptive feelings in your body as well so uh I think there is not just one attention mechanism I think attention is also very multi-level process and it's not only the kind of attention that we like to think of as um conscious mental action that we also can do on this kind of uh agent uh personal level but I think there are many uh subpersonal uh uh aspects also playing a big role in how attention then uh forms the kind of attentional experience that we have yeah yeah precisely precisely EXA exactly um yeah attent I've been trying to stress in my own internal dialogue and also with other people that I see Precision Precision waiting and attention is three different things they are not the same thing right Precision is a very technical mathematical IAL inverse entropy Precision waiting is in part informed by Precision but also context then and then attention is this kind of optimization of that Precision waiting over one tensor so I think that's kind of useful uh for people because most people are just used to hearing well attention is precision waiting it's like that won't do um I guess it even like confound this problem even more you have you start having stuff like priors over prior so you have like okay I have a prior about the actual distribution of my prior but I think the way that sanre explained this to me which is really useful is look at the brain and look how many ridges there are in the brain right like once you start actually folding in these hierarchies it all kind of makes sense um why the brain looks like that so and this is an interesting question I guess because of your expertise in in neuroscience and neuroanatomy do you think so we've now brought like four different priors into the game do you think that these are going to look the same uh and I know that's a very broad brush question but what I mean when I say look the same is are they going to have the same signature of neuronal firing are they going to be is the is the neurotransmitter associated with them going to be the same or are we just giving a semantic blanket to simplify the situation for something which is going they're going to be wildly different uh that's that's a very good point I mean the model that we are proposing this is a kind of functional model that we're seeing we we we are not yet sure uh how it will be uh implemented on a neural level but what we can say is um I mean a lot uh uh a lot of the things moving forward will will have something to do with uh dopaminergic Innovation for example but also uh other neuromodulators I think neuromodulators will play a big role in uh understanding uh this Precision model leting but not only for inference but also for for learning so um um I very much uh like Chris matis uh idea of precision weighted belief updating uh this is a context that he uh is a concept that he uses very frequently in in his research and um so one prediction would be if um you have no way of uh modulating uh top down uh Precision then toic issues keep going then you're also not updating your your prior beliefs right if it's if it's just a very fixed uh uh belief then no way of modulating it by dopamine for example you're stuck and this is something that uh actually uh might take place in uh in the Centro medial Mula where you have uh less flexibility uh there's less Innovation by dopaminergic regions and this is something that uh is uh yeah is is now uh uh yeah basically also of interest for uh for the effects of serotonergic uh medication that or even psychedelics that uh um neuroplasticity might be uh actually uh increased by by giving large doses of of Serotonin and this would then allow for re-update U updating uh those those preference priers in in those areas yeah yes okay in that picture it sounds like the pre preference priors or homeostatic prior you're want to call them are higher up than in some ways the perceptual priors yeah um okay that's really interesting because as I said before to have this flexible updating of your prior in terms of the perceptual belief updating you have to downregulate your Precision waiting over that PRI that perceptual prior right in turn uh increase the Precision over the likelihood distribution of the sensation but that seems to me to be governed at the higher level by an action policy right like do I want to relinquish Precision over this prior and maybe in many ways you know we have this idea of Rebus um which is

Carl and robing Carl Harris's notion that psychedelics flatten out the PRI landscape but that's it's a very very neat narrative but which priors right like all of them at which level and I think what uh what I'm really excited about in being an active inference in the community is this notion of deep parametric modeling the idea that we can't just take one slice of the hierarchy and say that that's sufficient to account for perception or action right because it it it runs deep unfortunately um I wonder yeah I wonder to kind of go to your teaching when you say okay this is a hierarchy and whatever you think is going on at like the perceptual level is going to be informed by by the conceptual level which is going to and God knows what's at the top how many this is a very stupid question but how many layers do you kind of envisage mentally when you're talking about that how and and if you can even say where does it stop what's the kind of guiding principle so it's definitely 7.2 I've heard people you I've heard I've heard people say seven I'm not even joking I've heard people say seven really yeah yeah seven or eight or maybe 12 no I mean I now I'm taking on my data scientist had actually uh if you add uh layers to a hierarchical model at some point they will run out of data to actually process so you will see that some uh layers will not process anything so these can gone can be gone and that's that's what the nervous system is actually doing so it's it's removing complexity whenever it can this is Beau things of the fre principle so if there's nothing to process they will be gone and I mean this has a lot in uh this also nicely links for example to uh the concept of uh of meditation where you say okay you basically dis you want to disengage with your normal way of being in the world of you know acting on the world having uh um goals in mind that you want to achieve but the only thing that you want to do is just sit there and be there in this very moment so what you're actually doing you create a state where not pro actively processing anything and you try to you know just calm everything down that you see and what is the experience that you're going to have if you have never meditated before yeah yeah well yeah it's just distraction it's distraction and and the the the the the effective feeling would be boredom right right so this is some way actually you don't have anything to process at this kind of level you you feel it une you don't know what to kind of do with this kind of experience that you're having but over over long time when you're trained to that um you will basically uh yeah get uh maybe uh uh um also this kind of hierarchical layer that would be necessary to actually process this kind of meter meter that's really nice kind of informative data and then you get used to it and you're fine and this is how you can add yet another uh level of the hierarchy oh that's really lovely yeah that's cool uh it's kind of like we're inventing our or or creating our own Markoff blankets our intermediate markof blankets between just I mean which we have right already I mean of course we do we're deeply hierarchical beings but I like the idea that we're kind of we're con it's context contingent in the sense I you know I think you mentioned actually in your paper that the central amigdala was kind of generally seen as responding to these kind of hardwired appetitive or aversive stimuli right so a tiger or a mango but actually that hasn't been fleshed out at all in experimental contexts because stimulus stimuli have subjective values right it's we don't all sit down and agree with God that we want mangoes and we don't want Tigers because maybe you have a pet tiger and maybe you're allergic to mangoes so maybe it's worth I don't know if you've got anything to say on that but like as you say there are clearly some hardw ones that we need to be human right to be this thing that persists through time that minimizes variational free energy but there are also others which are Downstream on subjective evaluation where do you kind of do you see those as two distinct categories with a bright line between them or do you see maybe the subjective evaluative ones emerging from the homeostatic ones where where's the what's the kind of interplay between those two yeah always um when whenever I start to think about these things it always looks a little bit like a a gradient uh so there there I think there are no exact boundaries point but I mean some of the very fixed beliefs we have about how the world should be I mean they feel very much real they feel like uh we're losing parts of ourself of ourselves if we give up this um I don't know this fact that we uh seem to believe about the world so we feel that everything falls down if we have to give it up even even though it's just something constructed even so even if it's just something made up right so um in a sense you could say that certain uh um beliefs uh uh have had adaptive value where you say this might might be also cultural norms that were uh transmitted and and so um there might be some some Merit to it um but uh um I think it's uh this uh there's there there still this gradient of things that you can easily unlearn and uh and and easily update and others where you uh we holding on a little bit harder yeah yeah yeah yeah cool so now we've been talking about hierarchies and the complexity of the human system and its capacity to adapt you also mentioned in brief in your paper the the possibility of engrs in the amydala so people associate engrs I mean you'll do a better job at explaining engrams than I will but with the hippocampus and with memory um and the way that I read that in the computational sense is kind of almost automated behavior um and habits and really you know strong priors over uh action so maybe it's worth starting there with the neuroanatomy because the amigdala is connected that has a route to the hippocampus and then say exactly what you mean because it sounds like oh the Amiga has a memory in some sense what does that mean and what does that how does that add to this already uh complicated story yeah what I like to um stress because I think this is really important is that I don't think that there are um that there are distinct memory and processing networks in the brain I think when you look at the and you could say uh I mean when you use the computer metaphor it's uh it's both it's both the CPU and the memory so one neuron already does this kind of computation and it me it has memory right so uh sometimes you know when we when we write uh cognitive Neuroscience textbooks we like to put big section headlines and we call one section uh memory and the other one decision making for example this is already what a neuron is doing it has its thresholds and it has its memories and the interesting question now is um what does a memory recognize actually and um uh some of the features or some of the neural computational role neuron is playing is only uh played when it's actually um when it's spiking when it shows activity but there is another aspect a neuron also um can be totally inactive and already plays a role in that by for example uh suppressing inputs that coming in it filters away noise and this is would be a very energy efficient way of actually uh doing neural processing when uh when it just shuts down things that are not needed anymore and only propagates those things that are meaningful deviations and then news worthy for the other parts of the hierarchy and this is what I'm actually mean with uh memory engrams so what what would that look like in terms of because for me and I make reference here to this new paper by George Dean and lar uh SED Smith and others looking at canalized Behavior it sounds to me that actually having kind of engrams or or or kind of a Proto memory system or a memory system in the amigdala could be somewhat uh pathological in the sense that if it's done you know non if it's a non-adaptive form of response then we have this habituated response that we have no prefrontal cortex regulation over that's freaking out every time we see um a rope right because I'm like oh I think that's a snake or a fa or a face in a rock because I have social anxiety so what what what's the Adaptive role of these kind of uh engrams and yeah what what's their that being so what's their role also potentially in something like psychopathology I mean I kind of see myself as also following this idea that uh brought forward by comput uh competitional Psychiatry people like Clan and car friston r Dolan uh and also Phil Colette of course because sometimes they they like to uh reframe pathologies in a way to say okay Hey listen people actually it's coming from a um there's some kind of good intention behind this mechanism optim there's actually an Adaptive value of depression of anxiety and so on and so on and in this context I would like to say yeah I mean it is good to be afraid of certain things and it's very good to have this kind of stereotypical uh reactions those habitual reaction to a stimulus that is coming if something is Dashing towards me and I will just you know raise my arms to protect my face then this is a very good adaptive response even though I mean even it might be a train uh running towards me and of course my hands will be um quite futile to useless yeah good luck but I mean it's the last resort that I have it's the only thing that I could potentially do and maybe it helps me but of course in the modern world it does not the the thing is uh many of these things pretty adaptive in in nature uh but of course something can go wrong um and the question is uh um is it always that is it actually going wrong or is it something that the brain cannot just no longer adapt to when you think of um of uh of this preference price many of them in the Micha I think are acquired during early parts of your childhood I mean this is studied in Mouse models and there is um indirect evidence also in humans for that so you basically learn uh what kind of place to world is whether it's a very dangerous place or whether it's a safe place and uh depending on that uh I mean the whole architecture of of the brain down to the biochemical level will change so I Mean Gene regulation is a big thing and uh the way the DNA is called up around the histone so How likely it is to for example uh produce stress hormones uh the these are things that are already uh um some some kind of not determined but influenced already at this early uh uh Parts in in in in in in development and developmental Windows where those things can actually uh uh take place and then they are there for the rest of your life in a way if not treated properly if not unlearned in a way so the thing is they might have had an advantage back then you know that the world is dangerous so you should be uh fast with your stress responses and uh so this could be uh then also the reason for many psychopathologies later on so some some people then uh will um find when they are grown up um different behavioral Solutions so some could be um yeah they are more anxiety prone because they know they are always basically the victim of a situation in a situation but other people might find that aggression is actually the goto solution for many things and um deligent Behavior can be a consequence from because they basically avoiding uh being hurt by hurting somebody else before that but the the the foundation for that could already be um basically uh uh um ingrained into those uh into those preference priors that are set early during um maturation yeah that's that's really fascinating because I think it adds another level of complexity to this kind of U bracketed category that we've been calling preference priors because I think reason reason my two sents on this is the reason why something becomes pathological as you say in many cases what we would consider pathological or maladaptive is very adaptive in certain contexts so in some ways it's the social context but it's also the capacity or the consequence of the individual reflecting on their own behavior and so things get worse you know in layman terms things get worse if you're like oh this is terrible like why can I not go to the party why can I not leave my house without checking the door 15 times and that seems to me that like what's What's Happening Here is that you've got two preference priors fighting with one another one is one is the higher order one I would actually say which is I kind of want to be a normal adaptive socially functioning human being and the other is uh there's a very high probability of there going to be danger and then these these both kind of feed into the perceptual Act and the active act so to speak and so you end up with this miscalibrated hierarchy where things are getting confirmed on one level which is clashing with the confirmation at another level which makes me think that there might be some truth in the idea that a sort of adapted human being is one where the the hierarchy is consonant or compatible the levels are compatible with each other all the way up so you don't have something like cognitive Discord or um yeah this kind of what I actually been calling opposition or self-modeling this kind of idea that we might be sustaining ourselves at one level but denying evidence at another level just wondering if you have any ideas about that and maybe how they actually feed into the way that we're perceiving and acting I know it's very complicated because it's hierarchical so it's always going to be complicated but I was wondering if you had any thoughts about that no I think this is a wonderful uh explanation that you gave already I I mean if we try to Now link it to to uh the am model uh I think it would be precisely this you I mean you have in anxiety disorders patients are aware that their anxiety their experience is excessive and it's no useful what they having but they cannot do anything about it so what they're feeling is a loss of control I mean first also about their bily states because um when you say social anxiety order they feel that the heart is Raising that they're sweating they Cann control the blushing whatsoever this loss of control actually feeds into their uh into their feel feeling of uh of threat of danger of Terror of uh being humiliated all those negative emotions so this is one thing the other thing is also the loss of control that they're not able to change this part of themselves so they cannot just update their belief to say okay but this is fine or this should be fine um and the reason could be um that what what we were saying uh um when we talked about the preference PRI you are not able to update them and for good reason because I mean you should uh you should learn uh during your early maturation what the world is like and you should stick to that because this is something that is apparently useful and you shouldn't just update it because you are in a good mood like you shouldn't change your preference prior for blood oxygen if something is interfering uh with with this kind of mechanisms like say opiates for example then uh the system might break and and you might die the same might also be for this kind of social preference prior of what social signals are dangerous and and um so it's very good that they work very efficiently without you paying attention to it in a very autonomic way and even if you're stupid even if you're drunk or whatever in this situation you shouldn't be able to just you know update those beliefs and and should be uh and bring yourself into into into danger great yeah I've got one more technical question and then then we we can leave it um I guess I can't really mentally figure out whether this is what's the kind of uh what's the kind of mechanism underlying this is it because on one hand it sounds like it's hyper or aberant Precision over the likelihood distribution and I think this maps onto something like hyperfixation or vigilance because the a is associated with attention and why I say that is okay so the the the likelihood is I would you I think it as why given X so the observation given the state and let's use our sort of if I've got a snake phobia if I say that if I put really high Precision over that likelihood M Matrix what I'm saying is these observations I've had is very likely to have come from the state of there being a snake MH at the same time I think of these disorders as being rooted in the priors I might have a hyper precise or extremely strong PRI that I can't be in the company of snakes and so if I get the slightest evidence of that given my likelihood distribution I'm my my posterior is so Divergent from my prior that that constitutes prediction error and a call to action so there are two kind of mechanisms going on there uh and maybe that has actually been UND depending my sort of General qualm with the with the notion of Prior and the ambiguity surrounding the word prior which I think has not been really discussed in the active inference community enough could you help us see what given from the perspective of the amydala is it both is it one of those is it neither like what's what's kind of going on here that's a fascinating question and and uh I mean you you are the first one to to ask me this kind of question and and I have to say I also thought about it I don't have a solution for it I mean it could have something to do with how we formulate the active inference model I think this problem um vanishes if we use a simpler hierarchical G and filter for example by Chris matis and uh I don't know maybe the active inance model um or C active inflence model got a a little bit overambitious at some point and and and is no longer pure in this very simple computational sense and and many many things were added to it to um yeah do many impressive things but maybe a more vanilla more simple model could actually also help us to to to figure it out uh so maybe there is no um no such a thing uh as a likelihood mapping uh that is different from uh from prior beliefs that we should care of oh but I don't know yeah well not yes I guess no one know I mean I the the the proof will be in the the experimental work and the the triangulation that can come from experimental work and the neurology and the neuros scanning and the computational modeling right um I think I think what you show very nicely here is that there's a limit to fear Rising um and I think I to be honest my perspective is active inference is really in its kind of uh it's at its highest moment of strength when it's extremely General so when I say I'm minimizing variational free energy all the time yeah like I'm 100% on board with that right like and and for me that that way that decomposes into perceptual uh adequacy and like normative adequacy is is really nice and the expected free energy equation is really nice but I guess these questions is well you're going to need some deep modeling here I guess these slight problem is this is stuff I've run into when I've been trying to think about these base graphs and modeling the phenomena that I've interested in there might be two base graphs which give you the same output um so which one wins right is it just Elegance because we want non-c complex but accurate models is it generalizability but you know so these are open questions but I think they're really important to raise in this kind of forum I I totally agree and this was also the motivation for this kind of uh paper because at one point we want to make uh empirical uh predictions I mean what we what we're doing in this uh in this paper at the moment is WE Post talk uh re frame uh uh certain explanations that we have in the literature and I mean they're all compatible with with with the story that we are telling it's just a new new way of looking at these things uh but the next step would be of course we have to I mean we're already making some kind of empirical predictions the next will be more precise computational uh uh predictions where we really look at uh how it could be what kind of modeling parameters that we use are actually reflected by some changes in the brain or behavior that we can actually observe uh with the methods that we have available and then only then we will see if this model is actually useful or if it's just a a nice folk story or something with no relevance for science whatsoever so this is what we have to find out and I mean this is also what I tell uh my my my colleagues from cognitive Neuroscience uh who engage now also in active inference and I tell them yeah we have to make good predictions empirical predictions not yet another modeling paper where uh where we show show the behavior of some artificial agent doing something because nowadays we come to the point that philosophers are making fun of us they say okay we are too theoretical we are too detached from really empirical practice and we should know get get back to the lab and and really try so this is definitely the road map for the future now first that I conceptualize this kind of um the the uh space of different phenomena that we usually like to think think of when we talk about Thea how it could actually map to a different narrative the next will be to come up with a computational model with uh precise predictions and then we are going to test it and I will yeah I haven't figured out what kind of architecture I actually um want to use fantastic fantastic well this is fascinating I mean the one thing we didn't even mention are the the parameters of some involved in perceptual updating and like the fact that they contribute um and here you can sort of have all your linear equations and stuff but we'll leave that we'll leave that to the mathematicians and I also don't think we're going to solve this problem today I mean there's there's an important thing that I want want to say as a a finisher for this for this last uh uh block because I mean I said I was so unsure about the laon Bing and and how the pricee actually sure uh uh encoded so I mean there's this idea of you know Cunningham's laws you you get an answer to your question by uh posting the the wrong answer to uh to a question on the internet and this is what I'm doing now I say okay no no no there is no such a thing as a likelihood mapping now go ahead and and show me how it's really done and the is also applies to this kind of uh new amca model I don't know if it's the right model but I I think we we want wanted to do something a little bit ambitious because what we found was that the typical narrative doesn't add up with what we observe in the human but also in the rodent data so I was collaborating with two mouse researchers and I mean when I talked to them about my ideas of effective inflence said yeah this might be the only way our data could make sense so there is some some problem with the existing theory that we having and maybe um we provided the wrong answer but I think if somebody's able to falsify us to to show us where we wrong I think we will learn a lot of what the Amal is actually doing this will be what the next step so yeah excellent well I I I appreciate both the both the pessimism and the optimism um um yeah and and sometimes I just wonder whether it's just we're just in a semantic cloud of confusion you know whether if we just can have different word like I've been trying to integrate this word autonomous prior and differentiate it from something like a a ceptual prior or hidden State prior and and then I you know again as I said we need to fold in stuff like parameters the parameters of our generative model and linear functions and the way that this might spit like the Y given X is really complicated especially in Dynamic stochastic worlds um it's not just like a static hidden varibles so hidden variables so you're absolutely right I think I think the money yes I mean there's this there's this um I think you're Absol right I mean I don't know whether humans have the capacity to model these things perfectly and maybe we don't need to um science doesn't necessarily work on perfect modeling it kind of works on effective predictions and so maybe we need to get back to the lab and do a bit more oddly go back to a bit more of a coar grained World um where if things work things work I I I totally agree so uh because I I think there is no big Narrative of what what the world really is and there's not one model to to model the whole world I think uh and this is uh I think one of the natural consequences of active inurance is that we can only go for an instrumentalised interpretation of what is going on because anything we observe is always conditional on something it's conditional on our model but it's also conditional on our actions so yeah we have certain uh we have certain preferences we want uh we want to achieve certain things and I mean the minimal preference could be not to Die For example and already causes us to inquire and to act in in a specific way when we're interacting with the world so there's always this conditional dependency that we should uh uh take under consideration and there is no way of you know getting the whole joint distribution it's always conditional on something so it's always all kinds of data that we are sampling is sampled for a particular unit case because that's thing that's that's perfectly put that's exactly what I wanted to say but for the last hour and a half but way better said Ronnie this was this was amazing I mean I I you know I oddly expected us to spend more time talking about the amigdala and less time talking about active inference more generally but I think you're you're one of the you're one of genuinely you're one of the only people I know who is teaching active inference directly on the ground day by day and I think that is a Hu I mean there's huge kudos to you because I can imagine that's not easy especially with the fact that it's updating the whole time we have different schema that we're employing with chucking some ideas out retaining some simplifying complexifying so thanks for doing that rather you than me I guess I'm doing that in part here um will be much worse but yeah no thank you so much this was incredibly useful I again I leave all of these conversations often times just to touch more confused than when started but I think that's a good Socratic sign that we're making progress um I always ask our our guests where can people find you I'm sure people are going to have lots of questions the amigdalar is a kind of permanent object of Fascination for people so if they wanted to ask questions or find out about the courses you're running where can people reach out to you so my uh blue sky and Twitter handle is sweet nuron and uh you can also find my website sweet neuron doat yeah and I mean I also like to say that I really loved uh talking to you even forgot this is uh recorded here because I mean I got so much in my in my head space and and talking about those interesting things so yeah I I love this format I Love Your Way of of doing it and it was truly an inspiring and thought-provoking um discussion oh you're very kind you're very kind I think that might be the best send off I've had I'm gonna I'm gonna that's that's made my day uh excellent Ronnie it's been an absolute treat I've learned a lot I'm confused like last week I'm going to go take a nap um thank you so much for joining me again thank you for me and The Institute thanks for hosting this